

Giovanni MINUTO



PROFILE

Innovative and passionate **Algorithms Developer** with over 3 years of experience in the Machine Learning industry, currently completing a **Ph.D.** at La Sapienza University. Over the past year, I held **Consulting and R&D roles**, developing innovative ML solutions for real-world problems. I am now seeking new professional opportunities in Turin or remotely.

CONTACT DETAILS

@minutogiovanni@gmail.com
+39 373 5495 348
GitHub
My Portfolio for More Info
Google Scholar

PERSONAL INFORMATION

- Address: Turin, Italy
- Citizenship: Italian
- Year of birth: 1997
- Languages: English (C1), Italian (native)

SKILLS

- **Programming Languages:** Python, Bash, C/C++, MATLAB/Octave, LaTeX
- **Machine Learning & Data Science Libraries:** PyTorch, Scikit-Learn, TensorFlow, Keras, NumPy
- **Quantum Computing Libraries:** Qiskit, PennyLane, Cirq, Qadence
- **Development Tools & Utilities:** Git, MLFlow, Docker, Multiprocessing, CI/CD pipelines
- **Methodologies & Practices:** Agile/Scrum, DevOps

WORK EXPERIENCE

2024–2025

QUANTUM ALGORITHMS DEVELOPER — *Pasqal, Netherlands*

- Provided consultancy to LG Electronics, leveraging Physics-Informed Neural Networks (PINNs) to model and solve complex multi-physics problems.
- Led a dedicated project vertical from design to deployment.
- Developed and maintained multiple libraries for machine learning applications.

2023–2025

TEACHING ASSISTANT — *Polytechnic School of Genoa, Italy*

- Courses: *Fundamentals of Quantum Computing; Introduction to Quantum Information and Computation for Robotics.*

EDUCATION

2022–TODAY

PHD IN QUANTUM MACHINE LEARNING — *La Sapienza, Italy*

- Research focus: Quantum Machine Learning.

2020–2022

MASTER IN PHYSICS — *University of Genoa, Italy*

- Thesis: *Physics Simulations with Quantum Computers* (Grade: 110/110).

2016–2020

BACHELOR OF SCIENCE IN PHYSICS — *University of Genoa, Italy*

- Thesis: *Josephson junctions and superconducting materials* (Grade: 96/110).

PUBLICATIONS

2025

- *A Novel Approach to Reduce Derivative Costs in Variational Quantum Algorithms, (Journal of Physics A: Mathematical and Theoretical, IOP).*
- *Resource reduction for variational quantum algorithms by non-demolition measurements, (Preprint).*

2023

- *Quantum gradient evaluation through quantum non-demolition measurements, (European Physical Journal D, Springer).*

OPEN SOURCE LIBRARIES

QADENCE:

- A simple interface to build digital-analog quantum programs on neutral atom devices.

QUANTUM NON-DEMOLITION MEASUREMENT ALGORITHM:

- Library for the QNDM algorithm, supporting gradient computation in quantum circuits.

CONFERENCES

04/2024

- *Joint ICTP-WE Heraeus School and Conference on Frontiers at the Intersection of Quantum Simulation and Machine Learning, Trieste, Italy.*

06/2023

- *Topics in Modern Machine Learning DIBRIS, University of Genoa, Genoa.*

02/2023–04/2023

- *Quantum Information Theory 2023 ICMAT, Autonomous University of Madrid, Madrid, Spain.*